DAY 12 | EXCEL

My Data Analyst Journey

SUMIFS, COUNTIFS & Excel Errors

Day 12 of learning Excel in public.

SUM IFS #N/A

Swipe to learn with me >

SUMIFS: The Idea

Same as SUMIF, but with more conditions

Why

SUMIF has one condition. SUMIFS has as many as you need.

Example

A report with Supplier Names down the side and Months across the top. Each cell needs two conditions.

Syntax

=SUMIFS(sum_range, criteria_range1, criteria1, criteria_range2, criteria2, ...)

Big Difference

In SUMIFS the sum range comes FIRST. In SUMIF it comes last.

Set Up the Report

Master Data and Reporting

Master Data

One row per sale: Date, Month, Supplier Name, Product, Region, Sales...

Report Rows

Every Supplier Name listed once (unique).

Report Columns

Every Month listed once, like Oct, Nov, Dec, Jan... Sep.

Total Column

=SUM(B3:M3) adds all the months for one supplier.

Build SUMIFS Step by Step

Use the Function Arguments box

Start

Type =SUMIFS, press Tab, then Ctrl+A to open the Function Arguments box.

Sum_range

Master Data > Sales column. Press F4 to lock it.

Criteria 1

Criteria_range1 = Supplier Names in Master Data (F4). Criteria1 = supplier names in your report.

Criteria 2

Criteria_range2 = Months in Master Data (F4). Criteria2 = months in your report. Click OK.

The Real Formula

From the practice sheet

Cell B3

=SUMIFS('Master Data'!$K$2:$K$101, 'Master Data'!$D$2:$D$101, A3:A17, 'Master Data'!$C$2:$C$101, B2:M2)

Microsoft 365

One formula in B3 fills the whole table (it spills). No dragging needed.

Older Excel

Use mixed references $A3 and B$2, then drag right and down.

Mixed Reference

$A3 locks the supplier column. B$2 locks the month row.

AVERAGEIFS & COUNTIFS

Same pattern, new result

`=AVERAGEIFS(K:K,D:D,$A3,C:C,B$2)` **Average, 2 rules**

`=COUNTIFS(D:D,$A3,C:C,B$2)` **Count, 2 rules**

`=COUNTIF(B3:M3,">0")` **Months with sales**

`=IFERROR(AVERAGEIFS(...),0)` **0 if no sales**

AVERAGEIFS puts the average range first, just like SUMIFS. COUNTIFS has no sum range, only pairs of range and criteria. COUNT cannot take a condition, so use COUNTIF for ">0".

Report Tips

Make it clean and correct

Zeros as Dashes

Accounting number format shows 0 as a dash.

Avoid #DIV/0!

AVERAGEIFS fails when a supplier had no sales that month. Wrap it in IFERROR.

Totals

Row total =SUM(B3:M3). Column total =SUM(B3:B17).

Check Your Work

The grand total must equal =SUM(K2:K101) from Master Data.

Excel Errors

The seven classic ones

#DIV/0!	Divide by 0 or a blank
#VALUE!	Wrong data type
#NAME?	Wrong function name
#REF!	Deleted cell reference
#N/A	Value not available
#NUM!	Invalid number math
#NULL!	Wrong range intersection

Error Examples

What causes each one

`=10/0`	#DIV/0!
`="Ali"+5`	#VALUE!
`=SUMM(A1:A5)`	#NAME?
`=A1+B1`	#REF! (delete B)
`=VLOOKUP(100,A2:B5,2,FALSE)`	#N/A
`=SQRT(-1)`	#NUM!
`=SUM(A2:A3 C1:C3)`	#NULL!

More Errors You Will Meet

Two bonus ones

######

The column is too narrow (or a date is negative). Double-click the column border to widen it.

#SPILL!

A spilling formula, like the SUMIFS report, is blocked by data in its way. Clear the cells in its path.

Green Triangle

Excel flags a possible problem. Click it for options like Trace Error.

Fix Errors in Bulk

Replace all errors at once

Step 1

Select the range, copy it and paste values (Ctrl+Alt+V, then V). The errors become constants.

Step 2

Press Ctrl+G > Special > Constants.

Step 3

Untick Numbers, Text and Logicals. Keep only Errors > OK. Every error cell is highlighted.

Step 4

Type 0 (or any value) and press Ctrl+Enter. All errors are replaced.

Better still: prevent errors with IFERROR before they appear.

Handle Errors with Formulas

Prevent, do not just hide

`=IFERROR(10/0,0)`	**0**
`=IFERROR(A1/B1,"")`	**Blank, not error**
`=IF(B1=0,0,A1/B1)`	**Avoid #DIV/0! first**
`=ISERROR(A1)`	**TRUE or FALSE**

IFERROR hides every error, including real mistakes. Use IF to avoid a known problem like dividing by zero.

DAY 12 | EXCEL

Day 12 Shortcuts

Build reports faster

Function Arguments box	Ctrl + A
Accept function name	Tab
Lock a range ($)	F4
Go To / Go To Special	Ctrl + G
Fill all selected cells	Ctrl + Enter
Paste values only	Ctrl + Alt + V + V
Evaluate Formula	Alt + M + V
Select whole column	Ctrl + Space

Day 12 Cheat Sheet

All functions in one place

Multiple Conditions

SUMIFS, AVERAGEIFS, COUNTIFS

Min / Max by Condition

MINIFS, MAXIFS

Handle Errors

IFERROR, IFNA, ISERROR, IF

Error Types

#DIV/0!, #VALUE!, #NAME?, #REF!, #N/A, #NUM!, #NULL!, #SPILL!

Day 12
Done!

SUMIFS, AVERAGEIFS, COUNTIFS
and Excel errors.

Which Excel error do you meet the most?
Tell me in the comments.

Follow for Day 13

Like | Save | Repost to help another beginner

Data Analyst Journey | Excel